Congressional Trade Dataset Preparation

Methodology for Data Cleaning, Enrichment, and Labeling

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Project Objective & Overview

Objective: To construct a clean, labeled dataset from raw congressional financial disclosures to enable analysis of trading behavior and performance.

Workflow:

1. Ingest raw transaction data.
2. Enrich data with market benchmarks (S&P 500) and historical prices.
3. Engineer features, including various return metrics.
4. **Clean and standardize** the dataset to ensure quality.
5. **Construct a binary label** to classify trade performance.

Initial Data:

`v5_transactions.csv`

- Rows: 35,365
- Columns: 26

Final Data:

`v9_transactions.csv`

- Rows: 32,416
- Columns: 47

Core Task 1.1: Data Cleaning - Structural & Missing Data

A significant portion of the project involved cleaning inconsistent and missing data. A total of **2,949 rows** were removed during this process.

Key Cleaning Steps (Part 1):

- **Duplicate Removal:** Checked for and removed 0 duplicate rows, ensuring unique transaction entries.
- **Column Pruning:** Removed columns with high nullity or low analytical value:
  - `Short` (35,314 NaNs), `Description` (32,785 NaNs), `Subholding` (14,042 NaNs), `District`.
- **Critical Missing Data Handling:**
  - Dropped rows where the critical `Label_1M` was `NaN` (e.g., due to insufficient historical data).
  - Filled categorical NaNs in `Committees`, `Industry`, and `Sector` with 'Unknown'.

Core Task 1.2: Data Cleaning - Standardization

Continuing the data cleaning process, focusing on standardizing categorical and structural features.

Key Cleaning Steps (Part 2):

- **Transaction Standardization:**
  - Consolidated 'Sale (Full)' and 'Sale (Partial)' into a single 'Sale' category.
  - Removed 266 'Exchange' transactions, as they do not represent typical buy/sell signals.
- **Party Affiliation Standardization:** Mapped abbreviations (D, R, I) to full names (Democrat, Republican, Independent).
- **Trade Size Standardization:** Converted raw 'Trade_Size_USD' values into standardized categorical ranges (e.g., '1 - 1000', '1001 - 15000').
- **Unique Transaction IDs:** Assigned a unique 'ID' to each transaction for easier tracking and referencing.

Missing Value Analysis

Understanding and addressing missing values was crucial for data quality.

Initial Missing Values (Top 5):

- `Short`: 35,314
- `Comments`: 34,525
- `Description`: 32,785
- `Subholding`: 14,042
- `District`: 8,227

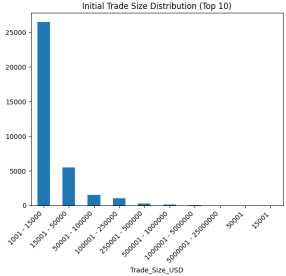
Final Missing Values (Post-Cleaning):

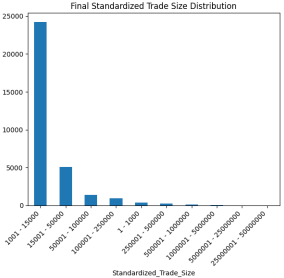
- `Comments`: 31,660 (Retained as non-critical for core analysis)
- `SPY_6Months`, `Label_6M`, `Excess_Return_6M`, `SPY_Return_6M`: ~1,539 (Due to future date projections or data availability)
- `SPY_3Months`, `Label_3M`, `Excess_Return_3M`, `SPY_Return_3M`: ~748 (Similar reasons as 6-month data)

The reduction in critical missing values ensures a robust dataset for modeling.

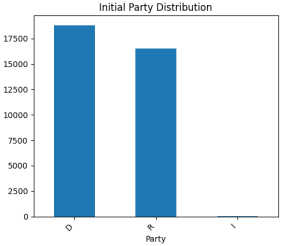
Visualizing Cleaning: Transaction & Party Standardization

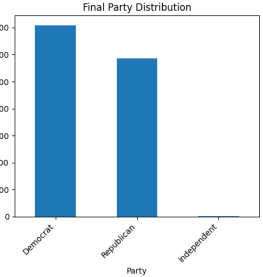
Transaction Consolidation





Party Affiliation Mapping





Core Task 2: Building the Trading Label

The primary goal is to create a binary label representing a "successful" trade. This requires defining success based on market outperformance.

Step 1: Calculate Excess Return

- First, we calculate the stock's return and the benchmark (SPY) return over a given period (1, 3, 6 months).
- Then, we define Excess Return as the difference:

$$\text{Excess Return} = R_{\text{stock}} - R_{\text{SPY}}$$

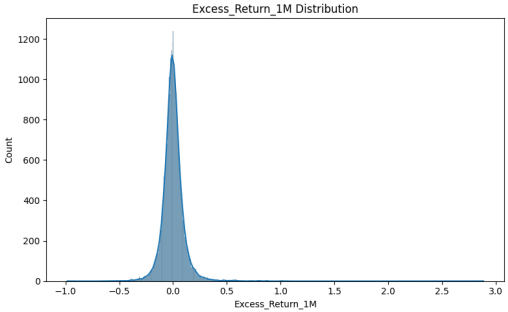
Step 2: Define the Label Logic

- We hypothesize a "good" trade outperforms the market by a significant margin, set here at **6%**.
- For a **Purchase**: Label $= 1$ if Excess Return > 0.06
- For a **Sale**: Label $= 1$ if Excess Return ≤ 0.06

Visualizing the Results: Returns and Labels

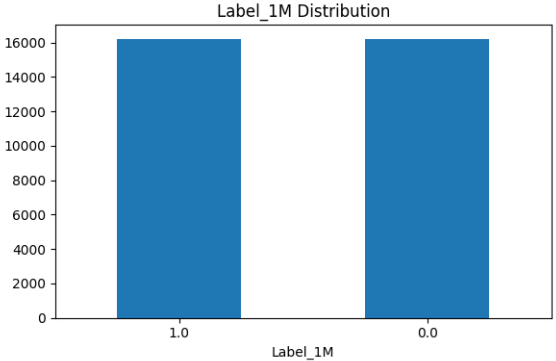
1-Month Excess Return Distribution

- The distribution is centered near zero, which is expected.
- It has long tails, indicating trades with significant outperformance or underperformance.



1-Month Label Distribution

- The resulting labels are almost perfectly balanced.
- This is ideal for training an unbiased machine learning model.



Conclusion & Next Steps

Accomplishments:

- A robust, automated pipeline for processing and cleaning financial disclosure data has been established.
- A clear, hypothesis-driven trading label (Label_1M) has been engineered, resulting in a well-balanced dataset.
- The final dataset is significantly enriched and standardized, making it suitable for advanced analysis.

Next Steps:

- Utilize this labeled dataset to train classification models (e.g., Logistic Regression, Gradient Boosting) to predict successful trades.
- Explore feature importance to understand what factors (e.g., party, committee assignments, sector) are most predictive of trade performance.
- Analyze the model's predictions to identify patterns in congressional trading.